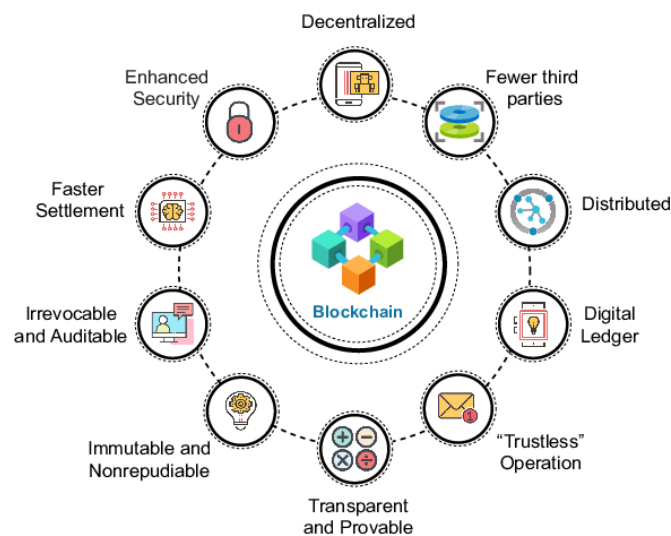


UNIT I

Blockchain is a decentralized digital ledger that records transactions across multiple computers, ensuring that once registered, transactions cannot be altered retroactively. This allows participants to confirm transactions without a central authority. While commonly used for cryptocurrencies like Bitcoin, blockchain technology also applies to supply chain management, voting systems, and smart contracts.



A **public ledger** is an open, decentralized database that records and verifies transactions in a transparent manner. These ledgers are accessible to anyone, allowing participants to view and confirm transactions without relying on a central authority.

Key Characteristics of Public Ledgers

1. **Transparency:** All transactions are visible to anyone, ensuring full transparency.
2. **Decentralization:** Operates on a network of distributed nodes, rather than a central server.
3. **Immutability:** Once recorded, transactions cannot be altered or deleted, ensuring the integrity of the data.
4. **Security:** Uses cryptographic techniques to secure transactions and prevent tampering.

Blockchain as Public Ledgers: The Case of Bitcoin

Blockchain technology serves as a public ledger for recording transactions in a decentralized and transparent manner. **Bitcoin**, the first and most well-known cryptocurrency, operates on a blockchain that exemplifies this concept.

How Bitcoin Uses Blockchain as a Public Ledger

1. **Transaction Recording:** Bitcoin transactions are recorded in blocks. Each block contains a list of transactions and a reference to the previous block, forming a chain.
2. **Decentralization:** The Bitcoin network consists of numerous nodes (computers) distributed globally. Each node maintains a copy of the entire blockchain, ensuring no single point of control or failure.
3. **Transparency:** All Bitcoin transactions are visible to anyone with internet access. This transparency allows participants to verify and audit transactions independently.
4. **Immutability:** Once a transaction is recorded in the Bitcoin blockchain, it cannot be altered or deleted. This immutability ensures the integrity and security of the transaction history.
5. **Security:** Bitcoin uses cryptographic techniques, such as hashing and digital signatures, to secure transactions and prevent tampering.

Benefits of Bitcoin's Public Ledger

- **Trustless Environment:** Participants do not need to trust a central authority; the system's design ensures trust through consensus mechanisms and cryptographic security.
- **Transparency:** Anyone can verify transactions, enhancing accountability and reducing the risk of fraud.
- **Security:** The decentralized and cryptographic nature of the blockchain provides robust protection against hacking and unauthorized changes.

Blockchain 2.0: The Evolution of Blockchain Technology

Blockchain 2.0 refers to the second generation of blockchain technology, expanding beyond cryptocurrency applications to include a wider range of

uses such as smart contracts, decentralized applications (dApps), and various other decentralized systems.

Key Features of Blockchain 2.0

1. **Smart Contracts:** Self-executing contracts with the terms of the agreement directly written into code. They automatically enforce and execute the contract when predefined conditions are met, eliminating the need for intermediaries.
2. **Decentralized Applications (dApps):** Applications that run on a blockchain network instead of a single computer. They are open-source, operate autonomously, and cannot be controlled by a single entity.
3. **Improved Scalability and Efficiency:** Blockchain 2.0 introduces enhancements in consensus mechanisms and data storage methods to handle a larger volume of transactions more efficiently.
4. **Interoperability:** Improved ability for different blockchain systems to communicate and work together, enabling seamless integration of various decentralized systems and applications.

Examples of Blockchain 2.0

- **Ethereum:** The most notable Blockchain 2.0 platform, Ethereum introduced the concept of smart contracts and provides a robust framework for developing decentralized applications.
- **Hyperledger Fabric:** A permissioned blockchain framework that supports smart contracts and is designed for enterprise use cases, focusing on enhanced privacy and performance.

Applications of Blockchain 2.0

1. **Finance and Banking:** Beyond cryptocurrencies, Blockchain 2.0 enables decentralized finance (DeFi) applications, which provide financial services like lending, borrowing, and trading without intermediaries.
2. **Supply Chain Management:** Enhances transparency and traceability of goods from production to delivery, ensuring authenticity and reducing fraud.
3. **Healthcare:** Secures patient records, enables efficient data sharing between providers, and ensures the integrity and privacy of sensitive information.

4. **Voting Systems:** Provides secure and transparent voting mechanisms, reducing the risk of fraud and ensuring the accuracy of election results.

Smart Contracts

Smart contracts are self-executing contracts with the terms of the agreement directly written into code. These contracts automatically enforce and execute actions when specific conditions are met, without the need for intermediaries like lawyers or notaries.

Key Features of Smart Contracts

1. **Automation:** Smart contracts automatically carry out the terms of the agreement when the predefined conditions are fulfilled. This reduces the need for manual intervention and speeds up processes.
2. **Transparency:** All parties involved can see the contract terms and conditions, ensuring clarity and trust.
3. **Security:** Smart contracts are stored on a blockchain, making them tamper-proof and secure from unauthorized changes.
4. **Efficiency:** By eliminating intermediaries and automating processes, smart contracts save time and reduce costs.

How Smart Contracts Work

1. **Code Creation:** The contract terms are translated into code, which is then stored on the blockchain.
2. **Condition Fulfillment:** When the specified conditions are met, the smart contract executes the agreed-upon actions automatically.
3. **Execution:** The contract actions (such as transferring money or property) are carried out and recorded on the blockchain, ensuring an immutable and transparent record.

Example of a Smart Contract

Imagine you are renting an apartment using a smart contract:

1. **Agreement:** The terms of the rental agreement (e.g., rent amount, rental period) are written into the smart contract.

2. **Condition:** The smart contract specifies that the rent must be paid by a certain date each month.
3. **Execution:** If the rent is paid on time, the smart contract automatically gives you access to the apartment. If the rent is not paid, the contract may automatically impose a penalty or terminate the agreement.

Benefits of Smart Contracts

- **Trust:** Eliminates the need for trust between parties by ensuring that the contract terms are automatically enforced.
- **Accuracy:** Reduces human error by precisely executing the contract terms as programmed.
- **Cost Savings:** Cuts out the middlemen, reducing fees and costs associated with traditional contract enforcement.

Distributed Consensus in Transactions

Distributed consensus is a process used in decentralized networks, like blockchains, to agree on the validity of transactions without a central authority. It ensures that all participants in the network have the same record of transactions.

Key Concepts of Distributed Consensus

1. **Decentralization:** No single entity controls the network. Instead, multiple participants (nodes) work together to verify and agree on transactions.
2. **Agreement:** All nodes in the network must agree on the state of the transactions to maintain a consistent and accurate ledger.
3. **Security:** Consensus mechanisms protect the network from fraud and malicious activities by making it difficult to alter the transaction history.

How Distributed Consensus Works

1. **Transaction Initiation:** A user initiates a transaction, such as transferring cryptocurrency from one account to another.

2. **Broadcasting:** The transaction is broadcast to the network, where it is visible to all participating nodes.
3. **Verification:** Nodes verify the transaction to ensure it is valid (e.g., the sender has enough balance, and the transaction follows the rules of the network).
4. **Consensus Mechanism:** Nodes use a consensus mechanism to agree on the validity of the transaction. Common consensus mechanisms include:
 - **Proof of Work (PoW):** Nodes solve complex mathematical puzzles to validate transactions. The first to solve the puzzle gets to add the block of transactions to the blockchain and is rewarded.
 - **Proof of Stake (PoS):** Nodes validate transactions based on the number of coins they hold and are willing to "stake" as collateral.
5. **Block Addition:** Once consensus is reached, the transaction is grouped with others into a block. The block is added to the blockchain, and the transaction is considered confirmed.
6. **Immutable Record:** The transaction is now part of the permanent and unchangeable blockchain ledger.

Example of Distributed Consensus in Bitcoin

1. **Transaction:** Alice wants to send 1 Bitcoin to Bob.
2. **Broadcasting:** Alice's transaction is broadcast to the Bitcoin network.
3. **Verification:** Nodes check that Alice has 1 Bitcoin to send and that the transaction is legitimate.
4. **Consensus (PoW):** Nodes (miners) compete to solve a puzzle. The first miner to solve it adds the transaction to a new block.
5. **Block Addition:** The new block, including Alice's transaction, is added to the blockchain.
6. **Confirmation:** The transaction is now confirmed, and Bob receives 1 Bitcoin.

Benefits of Distributed Consensus

- **Decentralization:** Removes the need for a central authority, distributing control across the network.
- **Security:** Makes it difficult for any single entity to alter the transaction history, protecting against fraud.

- **Transparency:** Ensures all participants have the same view of the transaction history, enhancing trust.

The Chain and the Longest Chain

The Chain:

In the world of cryptocurrency and blockchain technology, "the chain" refers to a blockchain. A blockchain is like a digital ledger that records transactions in a series of connected blocks. Each block contains a list of transactions, and once a block is full, it gets linked to the previous block, forming a chain of blocks - hence the name "blockchain."

Imagine a blockchain as a train, where each block is a train car filled with transaction data. As new transactions occur, they're added to the next available train car. Once that car is full, it's attached to the train, and the train keeps moving forward, adding new cars as needed.

The Longest Chain:

The concept of the longest chain is important in blockchain technology, especially in ensuring the integrity and security of the blockchain. In a decentralized network, multiple participants (called nodes) work on adding new blocks to the blockchain. Sometimes, two or more blocks might be created at the same time, leading to multiple chains.

The longest chain rule is a simple rule that helps resolve this situation. It states that the valid blockchain is the one with the most blocks, i.e., the longest chain. This is because the longest chain represents the most computational work done and is therefore considered the most secure and trustworthy.

So, if there's ever a disagreement on which chain is the correct one, the network will always choose the longest chain as the true blockchain.

Blockchain 2.0 - Permissioned Model of Blockchain

Blockchain 2.0:

Blockchain 2.0 refers to the evolution of blockchain technology beyond just recording financial transactions (like in Bitcoin) to more complex applications. This includes smart contracts, decentralized applications (dApps), and more.

Think of Blockchain 1.0 as a basic calculator that can only do simple math (like adding and subtracting), while Blockchain 2.0 is like a smartphone that can run various apps and perform many different tasks.

Permissioned Model of Blockchain:

In a permissioned blockchain, unlike a public blockchain (like Bitcoin), not everyone can join the network and validate transactions. Instead, there is a controlling entity or a group that decides who can participate.

Imagine a public park where anyone can enter and play (public blockchain) versus a private club where only members are allowed (permissioned blockchain).

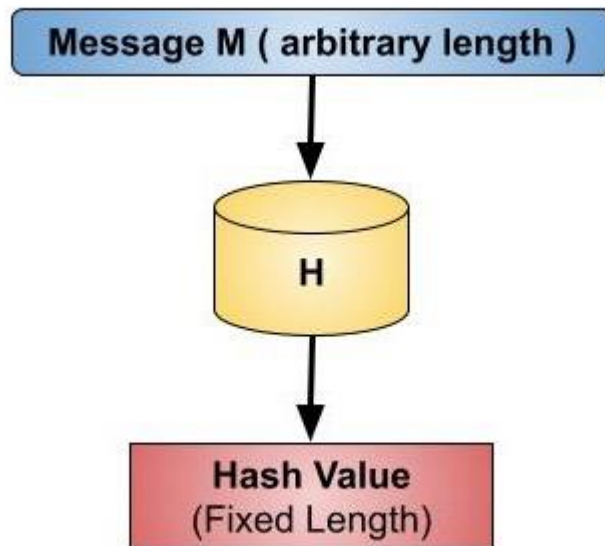
In a permissioned blockchain, participants need permission to join, and their identities are known to the network. This model is often used by businesses and organizations that require a higher level of control and privacy over their data. It ensures that only trusted parties can participate, which can enhance security and compliance with regulations.

Cryptographic Hash Function

Definition:

A cryptographic hash function is a special kind of mathematical function used in the world of cryptography (the science of secure communication). It takes an input (or "message") and produces a fixed-size string of

characters, which typically appears as a sequence of random letters and numbers. This output is called the "hash" or "digest."



Properties of a Cryptographic Hash Function

A cryptographic hash function has several important features that make it useful for securing data. Let's go through each feature one by one.

1. Deterministic:

- **Explanation:** The same input always produces the same output.
- **Example:** If you put "hello" into the hash function today, tomorrow, or any time in the future, you will always get the same hash result.

2. Fixed Size:

- **Explanation:** The output hash is always the same length, no matter how big or small the input is.
- **Example:** Whether you input a single word or an entire book, the hash might always be a 64-character string.

3. Fast Computation:

- **Explanation:** It takes a very short amount of time to compute the hash of any given input.
- **Example:** Even for large files, the hash function can quickly produce a hash.

4. Preimage Resistance:

- **Explanation:** Given a hash, it's extremely difficult to figure out what the original input was.
- **Example:** If you only have the hash "2cf24dba5fb0a30e26e83b2ac5b9e29e1b161e5c," it's practically impossible to determine that the original input was "hello."

5. **Collision Resistance:**

- **Explanation:** It is very unlikely that two different inputs will produce the same hash.
- **Example:** The hash of "hello" and the hash of "world" should be completely different. The chances of them being the same are extremely low.

6. **Avalanche Effect:**

- **Explanation:** A small change in the input causes a completely different hash.
- **Example:** If you change "hello" to "Hello" (note the capital 'H'), the hash will be entirely different, even though the inputs are very similar.

Hash Pointers in Blockchain

Hash pointers are a key element in the structure of blockchains.

How Hash Pointers Work in Blockchain

1. **Block Structure:**

- Each block in a blockchain contains a hash pointer to the previous block.
- This hash pointer is a combination of the address (pointer) and the hash of the previous block's content.

2. **Chain Integrity:**

- The hash pointer ensures that if any part of the previous block is altered, its hash will change.
- This change will propagate, invalidating the hash pointers in all subsequent blocks.

Benefits in Blockchain

- **Security:** Prevents tampering, as altering a block would require recalculating the hash for all subsequent blocks.
- **Immutability:** Ensures that the blockchain remains a secure, immutable ledger of transactions.

Example

- **Block 1:** Contains data and hash pointer to Block 0 (genesis block).
- **Block 2:** Contains data and hash pointer to Block 1.
- **Block 3:** Contains data and hash pointer to Block 2.

If an attacker changes data in Block 1, the hash pointer in Block 2 will not match the altered data, revealing the tampering.

Merkle Trees in Blockchain

Merkle trees are used to efficiently and securely verify data integrity within a block.

How Merkle Trees Work in Blockchain

1. **Transactions:**
 - Each transaction in a block is hashed to create leaf nodes of the Merkle tree.
2. **Tree Construction:**
 - Pairs of leaf nodes are hashed to form parent nodes, which are then paired and hashed again, continuing until a single root hash is formed.
3. **Root Hash:**
 - The root hash represents the entire set of transactions in the block.
 - It is stored in the block header, which also contains the hash pointer to the previous block.

Benefits in Blockchain

- **Efficient Verification:**
 - To verify a specific transaction, only a small number of hashes (Merkle path) need to be checked, rather than all transactions.

- **Scalability:**
 - Merkle trees enable the handling of large numbers of transactions efficiently.
- **Security:**
 - Ensures that any alteration in the transactions can be quickly detected by comparing the root hash.

Example: Merkle Tree with Four Transactions

Imagine we have four transactions, which we will call Tx1, Tx2, Tx3, and Tx4.

Step 1: Hash the Transactions

First, we calculate the hash of each transaction:

- $\text{Hash}(\text{Tx1}) = H1$
- $\text{Hash}(\text{Tx2}) = H2$
- $\text{Hash}(\text{Tx3}) = H3$
- $\text{Hash}(\text{Tx4}) = H4$

These hashes represent the leaf nodes of the Merkle tree.

Step 2: Create Parent Nodes

Next, we pair the hashes and hash them to create the parent nodes:

- $\text{Hash}(H1 + H2) = H12$
- $\text{Hash}(H3 + H4) = H34$

Here, + represents the concatenation of the two hashes.

Step 3: Create the Root Node

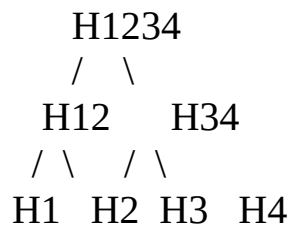
Finally, we hash the parent nodes to get the root node:

- $\text{Hash}(H12 + H34) = H1234$

This root hash represents the Merkle root of the tree.

Visual Representation

Here's a simple visual representation of the Merkle tree:



- **H1, H2, H3, H4:** Hashes of the individual transactions (leaf nodes).
- **H12:** Hash of H1 concatenated with H2.
- **H34:** Hash of H3 concatenated with H4.
- **H1234:** Root hash of H12 concatenated with H34.